

Baraa Wael

Backend Developer

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Artificial Intelligence & Computer Science undergraduate with experience in back-end and full-stack development using Node.js, Express.js, and MongoDB. Skilled in RESTful APIs design, role-based access control, and database optimization with practical exposure to real-time data processing and scalable architectures. Experienced in delivering projects across complaint management, computer vision-based exam monitoring, and airport surveillance systems. Certified in Big Data technologies with strong problem-solving, Git/GitHub, and collaborative development skills.

EDUCATION

Bachelor of Artificial Intelligence

Delta University for Science and Technology

GPA: 3.3/4.0

TECHNICAL SKILLS

- | | | |
|----------------------------------|---|---|
| • Node.js Development | • Express.js Framework | • MongoDB Database Design & Integration |
| • Data Modeling & Schema Design | • RESTful API Development | • Database Query Optimization |
| • Backend Architecture Design | • Role-Based Access Control (RBAC) | • Distributed Systems |
| • API Security & Data Validation | • Python, JavaScript | • Communication Skills |
| • Problem Solving | • Socket.io, Docker, Linux, CI/CD Pipelines | • Authentication & Authorization (JWT, OAuth) |
| • Machine Learning | • SQL & NoSQL Databases | • Leadership Skills |
| • Git & GitHub | • AWS Cloud | |

PROFESSIONAL EXPERIENCE

Freelance Web Application Developer

2024 – Present

- Develop and deliver custom database-driven web applications tailored to client requirements.
- Design and implement secure, scalable RESTful APIs to support dynamic front-end features.
- Integrate interactive dashboards for real-time data visualization and performance tracking.
- Optimize database queries and workflows to ensure efficient, seamless data flow across systems.
- Apply responsive design principles to deliver cross-platform, user-friendly web applications.

Backend Developer - Self-Learning Projects

2024 – Present

- Developed multiple back-end and full-stack applications leveraging Node.js, Express.js, and MongoDB.
- Designed and implemented RESTful APIs, integrated databases, and applied Role-Based Access Control (RBAC) across projects.
- Utilized Git and GitHub for version control, collaborative workflows, and project documentation.

Big Data Trainee - NTI

2025 – 08/2025

- Completed hands-on training in Big Data technologies with a focus on data processing, storage, and analytics.
- Gained practical experience with the Hadoop ecosystem, Spark, and distributed computing frameworks.
- Applied Big Data concepts to backend scenarios, enhancing proficiency in scalable, data-driven applications.
- Designed and executed data workflows for batch and real-time processing to simulate industry use cases.
- Strengthened problem-solving and technical skills by working on end-to-end data pipelines and system performance optimization.

Computer Vision Developer - Faculty of AI, Delta University

2024

- Engineered a real-time facial tracking and gaze analysis module to support secure online examination environments.
- Deployed the module into a web-based platform with integrated full-screen lockdown and automated reporting of suspicious activity.
- Designed and optimized computer vision pipelines with Python and OpenCV to enhance detection precision and reliability.
- Collaborated with a multidisciplinary team, strengthening expertise in version control and collaborative workflows using Git and GitHub.
- Gained practical experience in AI system deployment and seamless integration within production-level applications.

PROJECTS

MonlyKing (Gaming Accounts Marketplace) (Freelance Project)

08/2025 – 12/2025

Back-End Developer

A collaborative platform for buying and selling gaming accounts across popular titles including PUBG, Valorant, Fortnite, and FIFA. Developed in a two-person team, where I owned the entire back-end infrastructure while my partner handled the front-end.

- Designed and built **RESTful APIs** using **Node.js & Express** to handle all core platform operations including listings, orders, messaging, and payments
- Architected the **MongoDB** database schema with optimized indexes and well-structured collection relationships for efficient querying and filtering
- Implemented **Redis** caching layer to reduce database load and significantly improve response times for frequently accessed data
- Developed a full **role-based access control** system supporting three user types: Seller, Buyer, and Admin — each with distinct permissions and dashboards
- Built a real-time **internal messaging system** using [Socket.io](#), allowing direct communication between buyers and sellers linked to specific listings
- Integrated a complete **order and payment processing** system managing the full transaction lifecycle within the platform
- Configured a **Dockerized environment** with separate containers for Back-End, MongoDB, and Redis — streamlining deployment and ensuring environment consistency
- Collaborated closely with the front-end developer to ensure smooth **API integration** and consistent real-time data flow across the UI

Tech Stack: Node.js · Express.js · MongoDB · Redis · Docker · [Socket.io](#) · Git · GitHub

Elka2d — Integrated Educational Platform (Freelance Project)

09/2025 – 01/2026

Node.js & Express

A fully integrated online educational platform built for a single instructor to manage the entire learning process digitally. Developed in a 3-person team where I was solely responsible for all back-end development and system logic.

- Designed and built **RESTful APIs** using **Node.js & Express** covering Students, Courses, Lessons, Enrollments, Exams, and Attendance management
- Architected a structured **MySQL** relational database schema with well-defined relationships across all core entities
- Implemented a **Role-Based Access Control (RBAC)** system separating instructor and student permissions, ensuring each student can only access their enrolled content
- Built a **QR code-based attendance system** where each student is assigned a unique QR code — scanning it automatically logs attendance with date and time, while blocking unenrolled students
- Developed a full **examination system** allowing the instructor to create exams linked to specific courses and control which students are permitted to take them
- Designed the system with a **modular and scalable architecture** to support future growth and feature additions
- Collaborated closely with the front-end team on **API integration** and requirements analysis throughout the development process

Tech Stack: Node.js · Express.js · MySQL · React.js · Git · GitHub

Cheating Detection System (Team Project - Grade: A)

Python, OpenCV, Web Integration

- Led back-end development and integrated a computer vision module to detect real-time cheating behaviors during exam.
- Engineered optimized data pipelines with automated alert notifications for immediate incident reporting.
- Ensured robust back-end and front-end integration to enable seamless live monitoring and user interaction.

Airport Criminal & Fraud Detection System (Solo Project - Grade: A)

- Designed and implemented a facial recognition surveillance system to identify criminals and fraud suspects in airport environments.
- Developed a live camera streaming dashboard to monitor, track, and log entry/exit activity in real time.
- Built an interactive admin dashboard with event logs and real-time suspect tracking, delivering a production-ready full-stack solution with a robust back end, clean UI, and advanced computer vision capabilities.

Supermarket Management System with QR Code Support (Freelance Project)

- Developed a comprehensive system with multiple modules:
- VIP Customers: Managed customer profiles and applied automated loyalty discounts.
- Inventory Management: Tracked stock levels, suppliers, and product details.
- Sales: Recorded daily sales, applied discounts, and generated detailed reports.
- Cashier (POS): Built a user-friendly interface with QR Code scanning for faster checkout.
- Implemented role-based access control for admins, cashiers, and inventory managers.
- Integrated real-time database updates to ensure stock and sales accuracy.

Your Solution - Complaint Management System (Full-Stack Web App) (Freelance Project)

Backend: Node.js with Express.js | Frontend: Next.js | Database: MongoDB [LINK](#)

- Engineered a full-stack web solution to streamline complaint management in the hospitality industry.
- Implemented Role-Based Access Control (RBAC) to define permissions for customers, employees, and management.

- Designed an escalation workflow enabling complaint submission, tracking, and resolution.
- Integrated real-time notifications and dynamic status updates to improve responsiveness.
- Structured MongoDB with detailed history logs to ensure traceability and data integrity.
- Collaborated with a front-end developer who handled the UI, while I was solely responsible for all back-end development and API integration.

Database Systems for Hospital, Factory, and School (Desktop Applications)

MySQL, C#, Visual Studio

- Designed and deployed relational database systems to manage hospital, factory, and school operations efficiently.
- Built desktop applications with C# (Windows Forms) to provide intuitive user interfaces for staff and administrators.
- Optimized database schemas and validation logic, reducing data entry errors by ~30% and improving overall reliability.

CERTIFICATIONS

- Node.js, Express.js, and MongoDB Development
- Problem Solving & Logical Thinking Skills
- Big Data National Telecommunication Institute (NTI)

VOLUNTEERING

Public Relations Volunteer

2024 – Present

- Organize and promote student activities, workshops, and technical events while managing public relations to increase visibility and community engagement.
- Enhance communication, collaboration, and teamwork skills through active involvement in volunteer initiatives.

LANGUAGE

Arabic: Native • **English:** Professional Working Proficiency